

4)

$$g: \mathbb{R} \rightarrow \mathbb{R} : \text{Dom}(g) = \mathbb{R}$$

$$g(x) > 0 \quad \forall x \in \mathbb{R}$$

$$g(x) - x^2 - 25x > 0 \quad S = (-\infty; 27)$$

$$g(x) - x^2 - 25x < 0 \quad S = (27; +\infty)$$

Resolver:

$$\log_{\frac{7}{8}}(g(x) - x^2 - 25x) > \log_{\frac{7}{8}}(g(x))$$

$$\log_{\frac{7}{8}}(g(x) - x^2 - 25x) > \log_{\frac{7}{8}}(g(x))$$

$$\log_{\frac{7}{8}}(g(x) - x^2 - 25x) - \log_{\frac{7}{8}}(g(x)) > 0$$

$$\log_{\frac{7}{8}}\left(\frac{g(x) - x^2 - 25x}{g(x)}\right) > 0$$

$$\log_a(b) = c \iff a^c = b$$

$$\frac{7}{8}^0 > \frac{g(x) - x^2 - 25x}{g(x)}$$

$$1 > \frac{g(x) - x^2 - 25x}{g(x)}$$

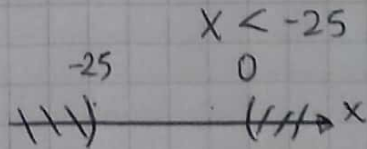
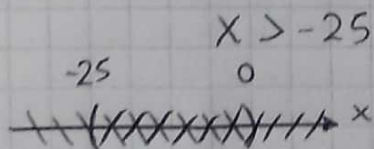
$$g(x) > g(x) - x^2 - 25x$$

$$0 > -x^2 - 25x$$

$$0 > x(-x - 25)$$

$$1) \quad x < 0 \wedge -x - 25 < 0$$

$$2) \quad x > 0 \wedge -x - 25 > 0$$



$$S = (-25; 0)$$